

PROJECT DOCUMENTATION

Visualization Tool for Electric Vehicle Charge and Range Analysis

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Domain: Data Analytics With Tableau

Project Title: Visualization Tool for Electric Vehicle Charge and Range Analysis

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INTRODUCTION

Electric Vehicles (EVs) are becoming increasingly popular worldwide due to concerns about pollution, rising fuel prices, and climate change. Governments and automobile manufacturers are investing heavily in EV technology and charging infrastructure. As EV adoption grows, a large amount of data is generated, including charging patterns, battery performance, energy consumption, driving range, and vehicle specifications.

Analyzing this data manually is difficult because it is stored across multiple datasets and contains numerous variables. Business Intelligence tools like Tableau help transform this raw data into visual dashboards, making it easier to identify trends and compare vehicle performance.

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** aims to simplify EV data analysis through interactive dashboards and stories. It enables users to compare EV models, analyze charging infrastructure, evaluate battery performance, and understand price differences. The project combines database management, data visualization, and web deployment to provide a complete analytics solution

PROBLEM STATEMENT

As electric vehicle adoption increases, organizations collect large amounts of data related to vehicle performance, charging stations, battery specifications, and driving range. Traditional methods of analyzing this information using spreadsheets or static reports are inefficient and do not provide interactive insights.

Consumers often find it difficult to compare EV models based on battery capacity, range, charging time, efficiency, and cost. Similarly, policymakers and researchers require visual tools to evaluate charging infrastructure and usage trends.

Therefore, there is a need for an interactive dashboard that transforms complex EV data into meaningful visual insights, enabling faster analysis and better decision-making.

OBJECTIVES

The primary objectives of the project are:

- To analyze electric vehicle charging behavior.
- To compare driving range among different EV models.
- To visualize charging station distribution across India.
- To compare battery capacity and charging time.
- To identify the most efficient EV brands.
- To analyze vehicle prices.
- To provide interactive filtering options.
- To create a responsive Tableau dashboard.
- To publish the dashboard using Flask.

Scope of the Project

This project can be used by:

- Electric Vehicle Manufacturers
- Government Transport Departments
- Charging Infrastructure Providers
- EV Buyers
- Automobile Researchers
- Students
- Data Analysts
- Environmental Organizations

The dashboard can also be expanded in the future by integrating real-time charging station data, battery health monitoring, and predictive analytics.

System Requirements

Hardware Requirements

- Intel Core i3/i5 Processor
- 4 GB RAM or higher
- 10 GB Free Disk Space
- Internet Connection

Software Requirements

- Windows 10/11
- MySQL Workbench
- Tableau Desktop
- Python 3.x
- Flask
- Visual Studio Code
- Google Chrome

Technologies Used

MySQL

MySQL is an open-source relational database management system used to store the electric vehicle dataset. It supports structured data storage, SQL querying, and efficient retrieval of records.

Role in the Project

- Store EV data
- Execute SQL queries
- Connect Tableau to the database
- Manage tables and records

Tableau

Tableau is a Business Intelligence tool used to create interactive charts, maps, dashboards, and stories.

Features Used

- Bar Charts
- Pie Charts
- Maps
- Filters
- Calculated Fields
- Dashboards
- Stories

Python

Python is used to run the Flask application and support backend operations required for deployment.

Flask

Flask is a lightweight Python web framework used to publish the Tableau dashboard on a web interface.

System Architecture

The system follows the workflow below:

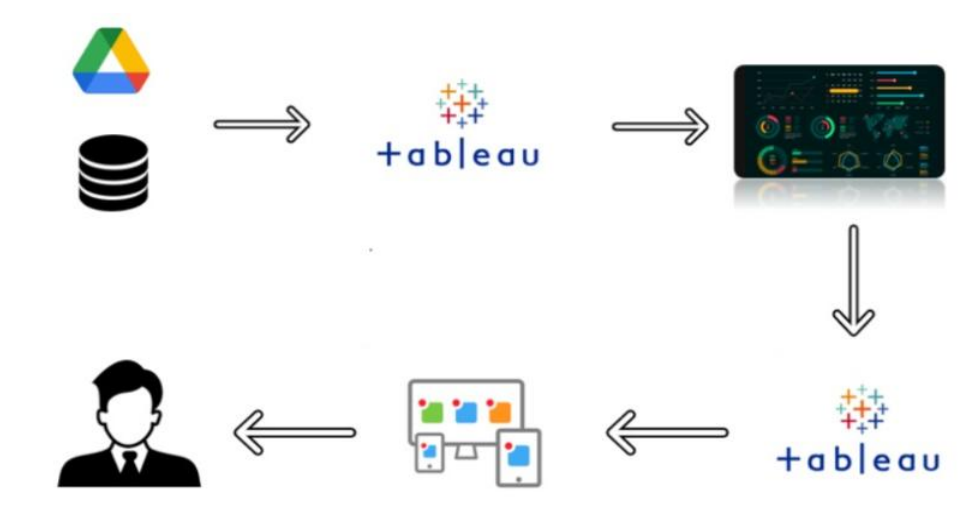
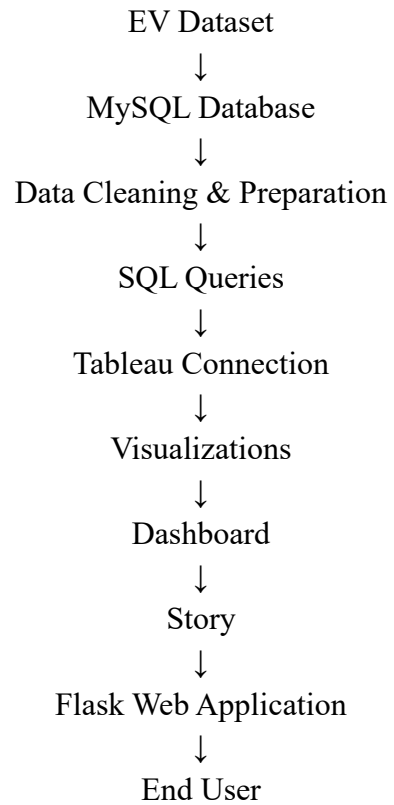


Figure 0.1: Architecture

STEP-1: Data Collection & Extraction from Database

i)Dataset Description

The dataset contains information about electric vehicles and charging infrastructure.

Main Attributes

- Vehicle Brand
- Vehicle Model
- Battery Capacity (kWh)
- Driving Range (km)
- Charging Time
- Charging Speed
- Top Speed
- Efficiency (Wh/km)
- Price
- Body Style
- Powertrain Type
- Charging Station Region
- Charging Station Type

Each attribute contributes to different analyses such as efficiency comparison, price analysis, charging behavior, and brand performance.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Brand	Model	AccelSec	TopSpeed_KmH	Range_Km	Efficiency_WhKm	FastCharge_KmH	RapidCharge	PowerTrain	PlugType	BodyStyle	Segment	Seats	PriceEuro				
1	Tesla	Model 3 Long Range Dual Motor	4.6	233	450	161	940	Yes	AWD	Type 2 CCS	Sedan	D	5	55480				
2	Volkswagen	ID.3 Pure	10	160	270	187	250	Yes	RWD	Type 2 CCS	Hatchback	C	5	30000				
3	Polestar	Polestar 2	4.7	210	400	181	620	Yes	AWD	Type 2 CCS	Liftback	D	5	56440				
4	BMW	iX3	6.8	180	360	206	560	Yes	RWD	Type 2 CCS	SUV	D	5	68040				
5	Honda	e	9.5	145	170	168	190	Yes	RWD	Type 2 CCS	Hatchback	B	4	32997				
6	Lucid	Air	2.8	250	610	180	620	Yes	AWD	Type 2 CCS	Sedan	F	5	105000				
7	Volkswagen	e-Golf	9.6	150	190	168	220	Yes	FWD	Type 2 CCS	Hatchback	C	5	31900				
8	Peugeot	e-208	8.1	150	275	164	420	Yes	FWD	Type 2 CCS	Hatchback	B	5	29682				
9	Tesla	Model 3 Standard Range Plus	5.6	225	310	153	650	Yes	RWD	Type 2 CCS	Sedan	D	5	46380				
10	Audi	Q4 e-tron	6.3	180	400	193	540	Yes	AWD	Type 2 CCS	SUV	D	5	55000				
11	Mercedes	EQC 400 4MATIC	5.1	180	370	216	440	Yes	AWD	Type 2 CCS	SUV	D	5	69484				
12	Nissan	Leaf	7.9	144	220	164	230	Yes	FWD	Type 2 CHAd	Hatchback	C	5	29234				
13	Hyundai	Kona Electric 64 kWh	7.9	167	400	160	380	Yes	FWD	Type 2 CCS	SUV	B	5	40795				
14	BMW	i4	4	200	450	178	650	Yes	RWD	Type 2 CCS	Sedan	D	5	65000				
15	Hyundai	IONIQ Electric	9.7	165	250	153	210	Yes	FWD	Type 2 CCS	Liftback	C	5	34459				
16	Volkswagen	ID.3 Pro S	7.9	160	440	175	590	Yes	RWD	Type 2 CCS	Hatchback	C	4	40936				
17	Porsche	Taycan Turbo S	2.8	260	375	223	780	Yes	AWD	Type 2 CCS	Sedan	F	4	180781				
18	Volkswagen	e-Up!	11.9	130	195	166	170	Yes	FWD	Type 2 CCS	Hatchback	A	4	21421				
19	MG	ZS EV	8.2	140	220	193	260	Yes	FWD	Type 2 CCS	SUV	B	5	30000				
20	Mini	Cooper SE	7.3	150	185	156	260	Yes	FWD	Type 2 CCS	Hatchback	B	4	31681				
21	Opel	Corsa-e	8.1	150	275	164	420	Yes	FWD	Type 2 CCS	Hatchback	B	5	29146				
22	Tesla	Model Y Long Range Dual Motor	5.1	217	425	171	930	Yes	AWD	Type 2 CCS	SUV	D	7	58620				
23	Skoda	Enyaq iV 50	10	160	290	179	230	Yes	RWD	Type 2 CCS	SUV	C	5	35000				
24	Audi	e-tron GT	3.5	240	425	197	850	Yes	AWD	Type 2 CCS	Sedan	F	4	125000				
25	Tesla	Model 3 Long Range Performance	3.4	261	435	167	910	Yes	AWD	Type 2 CCS	Sedan	D	5	61480				
26	Volkswagen	ID.4	7.5	160	420	183	560	Yes	RWD	Type 2 CCS	SUV	C	5	45000				
27	Volkswagen	ID.4	7.5	160	420	183	560	Yes	RWD	Type 2 CCS	SUV	C	5	45000				

Figure 1.1: Example Dataset

ii) Storing Data in DB & Perform SQL Operations

The EV dataset is imported into a MySQL database.

Steps

1. Create a new database.
2. Create the required table(s).
3. Import the dataset.
4. Verify data integrity.
5. Execute SQL queries for analysis.

Example query:

```
SELECT * FROM electric_vehicle;
```

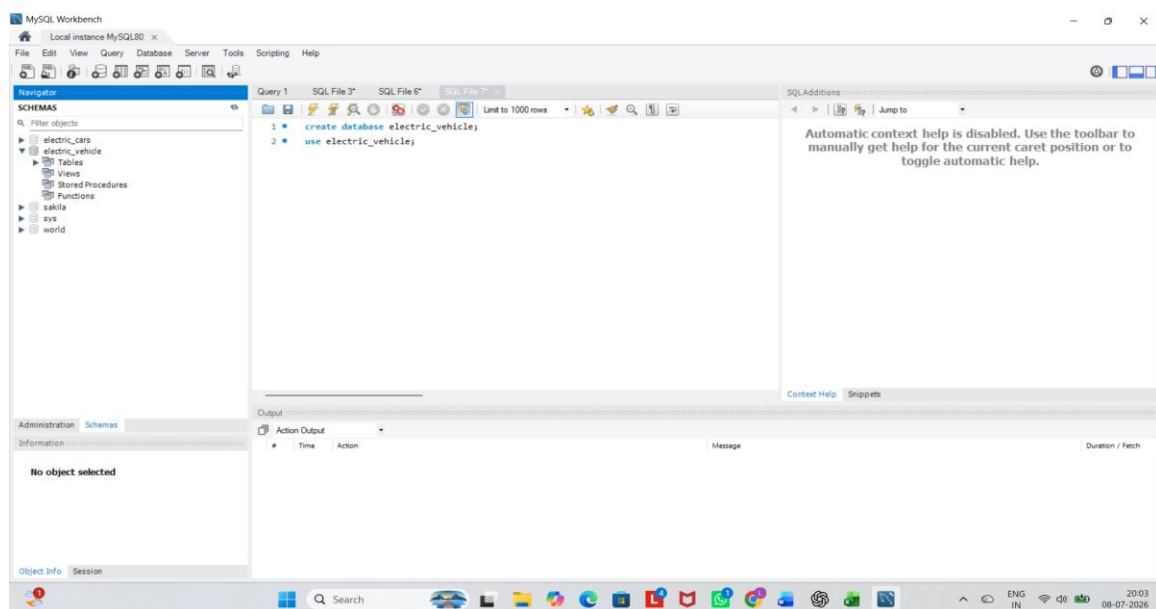


Figure 1.2: Creation of Database

iii)Connect DB with Tableau

Establishing a secure and reliable connection between the database and Tableau. Ensure real-time or scheduled data access for creating interactive dashboards and performing dynamic data analysis within the Tableau environment.

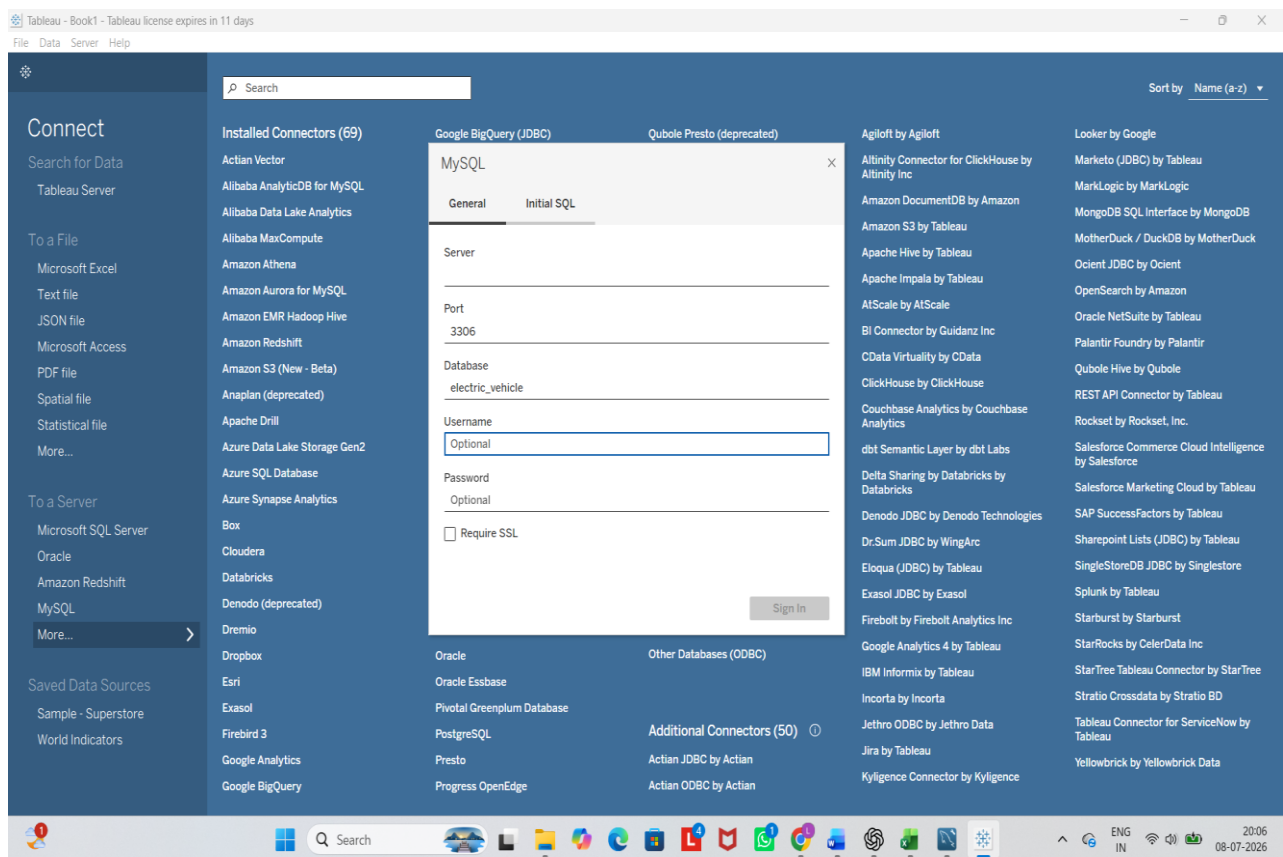


Figure 1.3: Connecting DB with Tableau

STEP-2: Data Preparation

Prepare the Data for Visualization

Performing Various operations on the data like:

- Data Review & Exploration
- Filtering and Structuring for Purpose
- Data Cleaning
- Data Validation
- Field renaming and final formatting
- Optional calculated fields

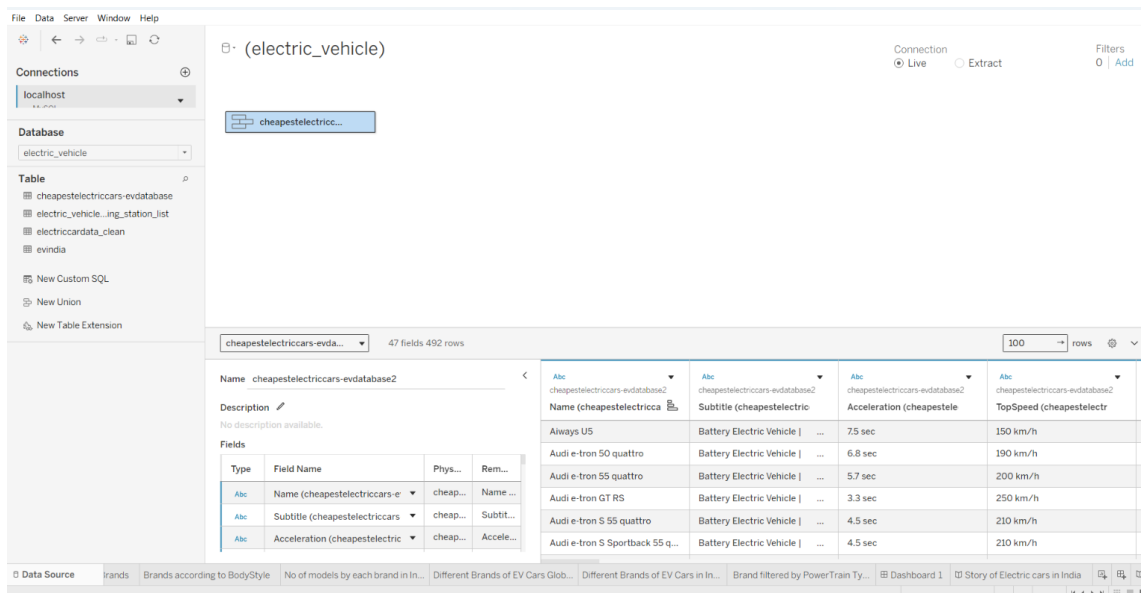


Figure 2.1: Preparing Dataset for Visualization

STEP-3: Data Visualization

Data visualization is the process of creating graphical representations of data in order to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

Examples include:

- Charging Stations by Region and Type in India
- EV Charging Stations Map of India
- Different EV Cars in India
- Top Speed for different Brand
- Price For different cars in India
- Top 10 Most Efficient EV Brands
- Brands according to Bodystyle
- Brand Filtered by Powertrain Type

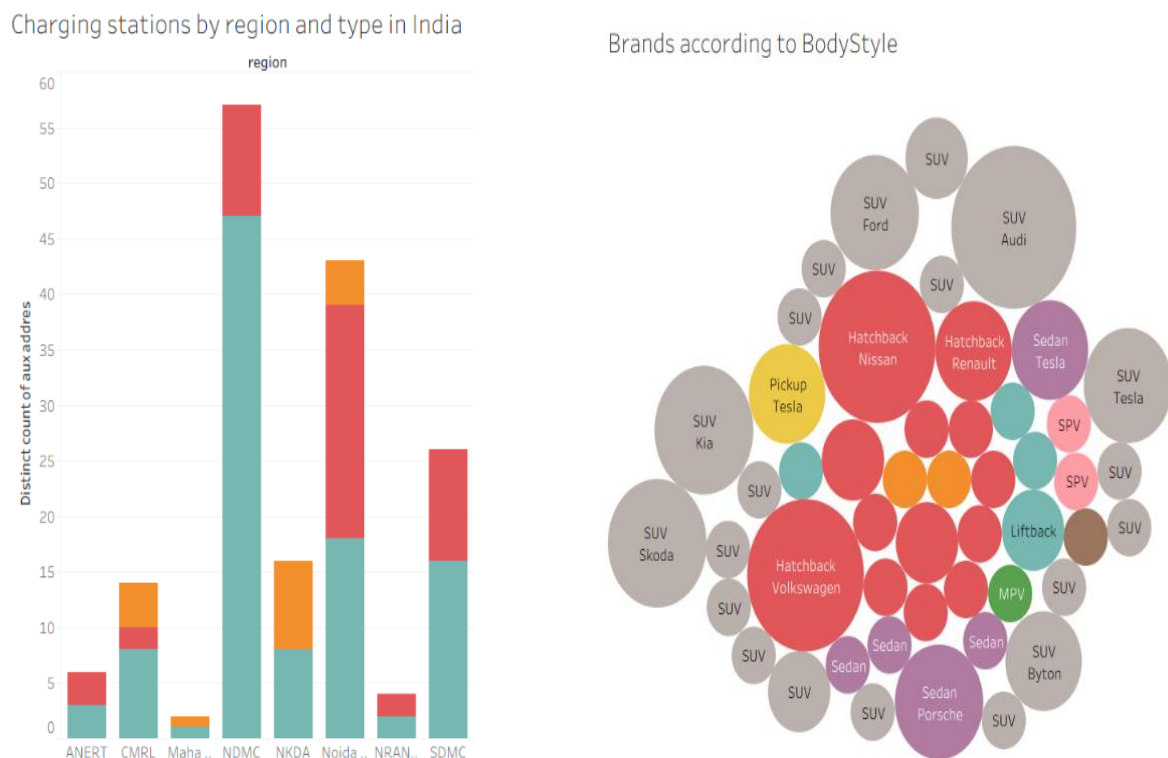


Figure 3.1: Example for Visualization

STEP-4: Dashboard

A **dashboard** is a graphical user interface (GUI) that displays data in an organized and easy-to-understand format. It combines multiple visualizations such as charts, graphs, maps, and tables into a single screen, allowing users to monitor key performance indicators (KPIs), analyze trends, compare data, and make informed decisions. Dashboards provide a clear overview of large datasets and enable users to interact with the data using filters and tooltips.

Responsive Dashboard

The dashboard developed for the **Visualization Tool for Electric Vehicle Charge and Range Analysis** is designed to be responsive and user-friendly. It adapts to different screen sizes, including desktops, tablets, and mobile devices, ensuring a consistent viewing experience. The dashboard combines multiple Tableau worksheets into a single interactive interface and includes visualizations such as charging station analysis, EV charging station map, top speed comparison, price comparison, battery efficiency, body style distribution, and powertrain analysis. Interactive filters and tooltips help users explore the data efficiently and gain meaningful insights into electric vehicle performance and charging infrastructure.

The interactive Tableau dashboard developed for this project can be accessed using the following link:

Dashboard URL:

https://public.tableau.com/app/profile/lakshmisetty.meghana/viz/ElectricCarAnalyticsDashboard_17832499846150/Dashboard1?publish=yes



STEP-5: Story

In Tableau, a scene (also called a story point) is an individual step within a Story. Each scene displays a specific worksheet, dashboard, or text explanation, focusing on a particular aspect of the data. Scenes guide users through the analysis in a logical sequence, making it easier to understand trends, compare insights, and draw meaningful conclusions.

For this project, multiple scenes were created to present different aspects of the Electric Vehicle Charge and Range Analysis. Each scene highlights a specific analysis, such as charging station distribution, battery performance, driving range, price comparison, and vehicle efficiency, providing users with a clear and interactive data storytelling experience.

The scenes created for this project are shown in the screenshot below.

Story of Electric cars in India

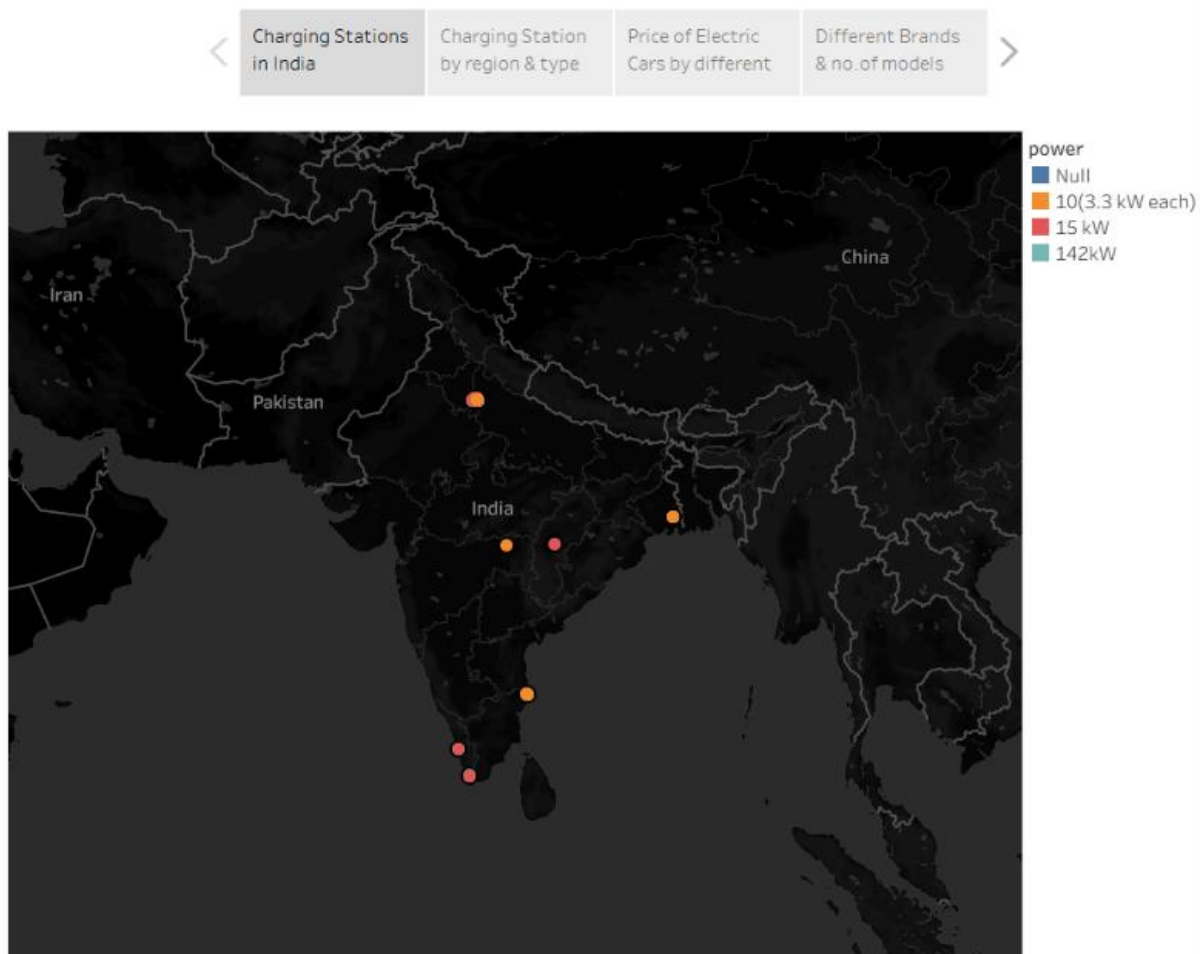


Figure 5.1: Scenes of Story

STEP-6: Performance Testing

Performance testing was carried out to evaluate the efficiency and responsiveness of the Electric Vehicle Charge and Range Analysis Dashboard. The testing focused on the amount of data retrieved from the database, the effectiveness of data filters, the use of calculated fields, and the number of visualizations displayed. Optimizing these factors ensures that the dashboard loads quickly, responds efficiently to user interactions, and provides accurate analytical insights.

i) Amount of Data Rendered from the Database

The EV dataset is stored in a MySQL database. During testing, the database was monitored to ensure that data retrieval was efficient and did not affect dashboard performance. Table information, including the number of rows and columns, was verified using MySQL Workbench, ensuring that the required data was successfully rendered in Tableau.

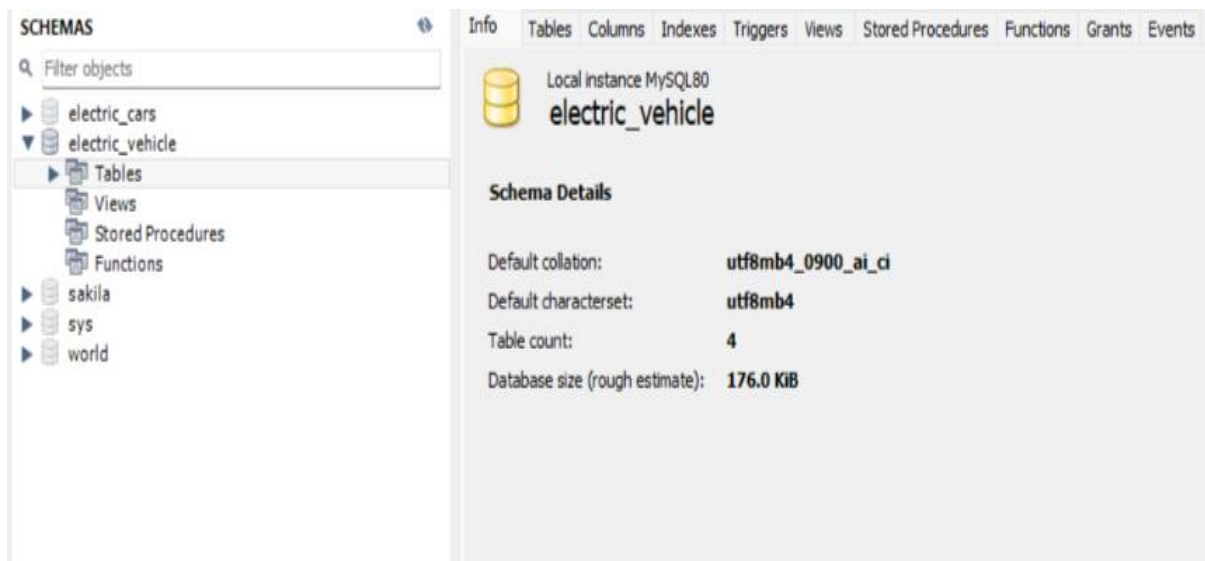


Figure 6.1: MySQL DB Schema and Table Details

ii) Utilization of Data Filters

Interactive filters were implemented to improve user experience and dashboard responsiveness by displaying only the required data.

The dashboard includes the following filters:

- Body Style Filter – Displays vehicles based on body style such as Hatchback, Sedan, SUV, MPV, Pickup, Cabrio, and Station Wagon.

- **Brand Filter** – Allows users to select specific EV manufacturers such as Tesla, Tata, BMW, Audi, Mercedes-Benz, Hyundai, Jaguar, Porsche, and others.
- **PowerTrain Filter** – Filters vehicles by drivetrain type, including AWD (All-Wheel Drive), FWD (Front-Wheel Drive), and RWD (Rear-Wheel Drive).
- **Car Filter** – Enables selection of individual EV models for detailed analysis.

These filters improve dashboard interactivity and help users analyze specific segments of the dataset efficiently.

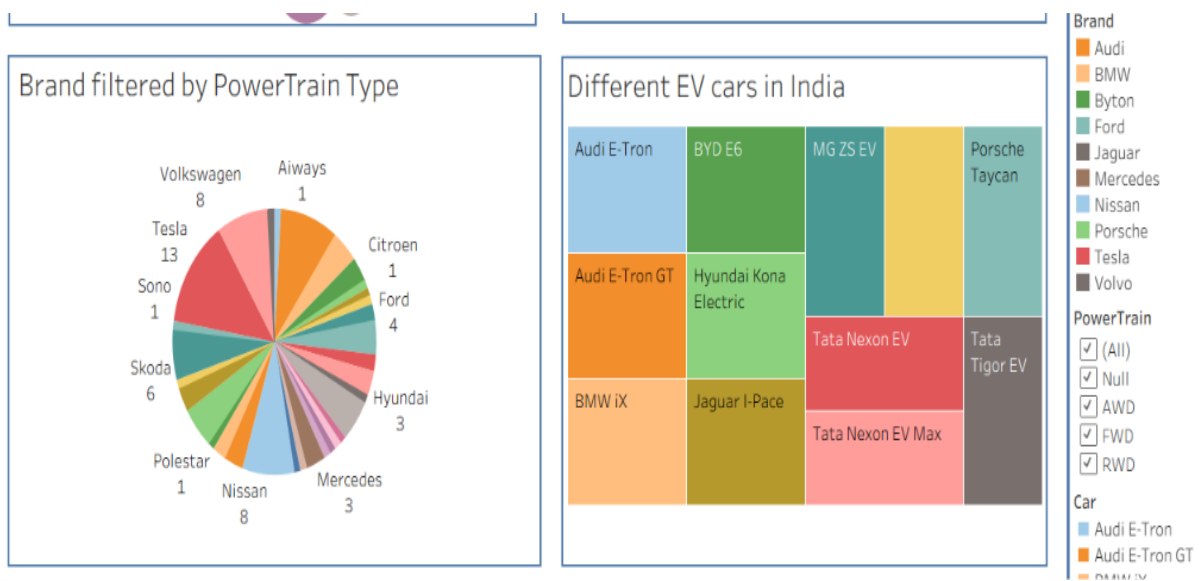


Figure 6.2: Visualization using Dashboard Filters

iii) Calculated Fields

Calculated fields were created in Tableau to perform additional analysis without modifying the original dataset. These fields provide summary statistics and support the visualizations.

Steps to Create Calculation Field:

- Right-click in the Data Pane (left side).
- Select Create Calculated Field.
- Type a name (example: Count).
- Write a formula (example: COUNT([Car])).
- Click OK.

The calculated fields used in this project include:

- **Count** – Calculates the total number of electric vehicle models.

- Count_PowerTrain – Calculates the total number of vehicles for each powertrain category.

These calculated fields improve the accuracy of charts and summary cards.

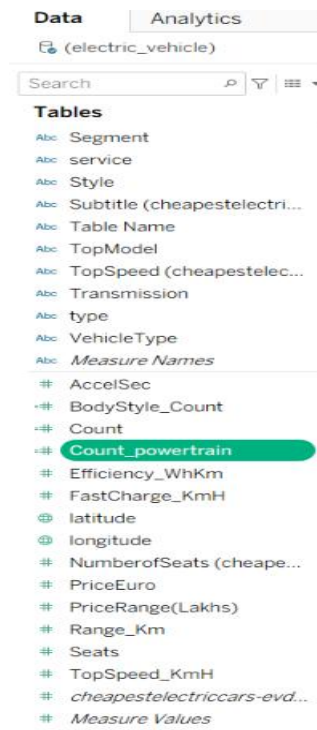


Figure 6.3: Creation of Calculated Fields

iv) Number of Visualizations

The dashboard includes multiple visualizations to provide comprehensive insights into electric vehicle data. The visualizations developed are:

- Charging Stations by Region and Type in India
- EV Charging Stations Map of India
- Different EV Cars in India
- Top Speed for Different Brands
- Price Comparison of Different EV Cars in India
- Top 10 Most Efficient EV Brands
- Brands According to Body Style
- Brand Filtered by PowerTrain Type
- Number of Models by Each Brand
- Summary Card for Different Brands of EV Cars Globally
- Summary Card for Different Brands of EV Cars in India

These visualizations present the data in an interactive and easy-to-understand format, enabling users to compare vehicle performance, charging infrastructure, efficiency, and pricing effectively.

STEP-7: Web Integration

i)Publishing

Publishing is the process of making Tableau dashboards and stories available online so that users can access and interact with them through a web browser. It enables users to view visualizations, analyze trends, monitor key performance indicators (KPIs), and share insights with others. Publishing also allows dashboards and stories to be embedded into websites or shared using a public link, making them accessible from anywhere.

In this project, both the **Electric Vehicle Charge and Range Analysis Dashboard** and the **Tableau Story** were published using **Tableau Public**. Before publishing, all worksheets, dashboards, stories, filters, calculated fields, and interactive features were verified to ensure they functioned correctly. After signing in to Tableau Public, the workbook was uploaded and made publicly accessible through the Tableau Public profile.

Publishing the dashboard and story provides several benefits, including easy accessibility, online sharing, collaboration, and secure storage. Users can interact with the dashboard to explore electric vehicle data through filters and visualizations, while the story presents the analysis in a logical sequence, helping users understand the insights step by step. The published workbook can be accessed through any web browser without requiring Tableau Desktop.

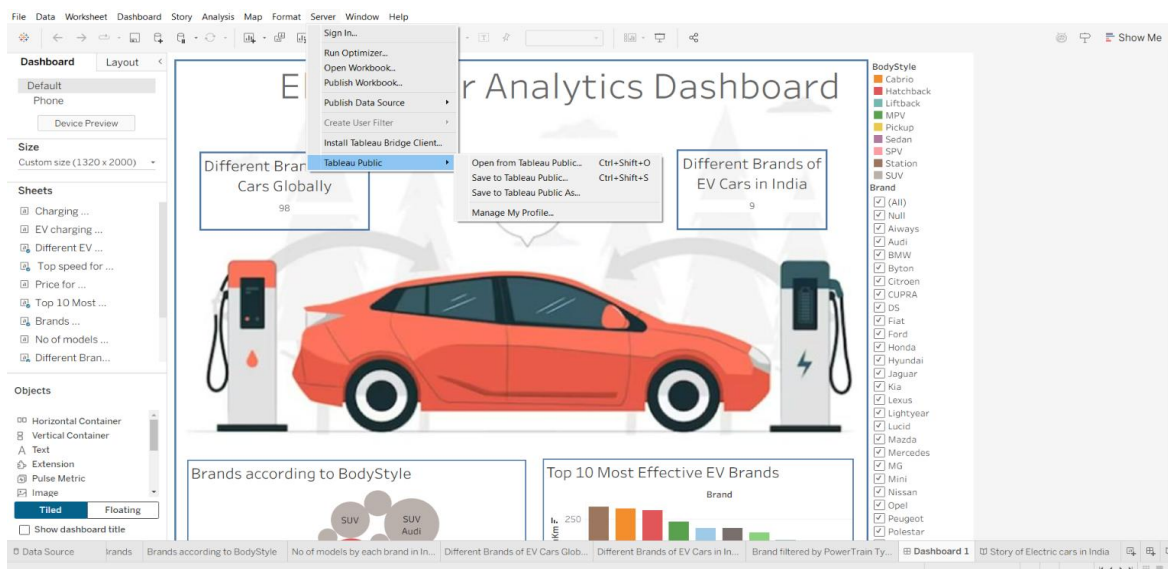


Figure 7.1: Publishing the Tableau Dashboard

Dashboard Link

Electric Vehicle Charge and Range Analysis Dashboard:

https://public.tableau.com/app/profile/lakshmisetty.meghana/viz/ElectricCarAnalyticsDashboard_17832499846150/Dashboard1?publish=yes

Story Link

https://public.tableau.com/views/ElectricCarAnalytics_17832693608520/StoryofElectriccarsinIndia

ii)Dashboard and Story Embed with UI Using Flask

Flask is a lightweight Python web framework used to develop web applications. In this project, Flask was used to create a simple and user-friendly web interface for displaying the **Electric Vehicle Charge and Range Analysis Dashboard** and **Tableau Story**. The published Tableau workbook was embedded into the Flask application, allowing users to access interactive visualizations directly through a web browser.

The web interface provides a centralized platform where users can easily navigate between the dashboard and the story. The embedded dashboard enables users to interact with filters, charts, maps, and other visualizations to analyze electric vehicle charging infrastructure, battery performance, driving range, efficiency, and pricing. The embedded story presents the analysis in a logical sequence, helping users understand the insights step by step.

Using Flask improves the accessibility and usability of the project by allowing users to view the dashboard and story without requiring Tableau Desktop. The web application is responsive, easy to use, and can be accessed from different devices such as desktops, laptops, tablets, and smartphones, making the project convenient to share and demonstrate.

Features of the Web Interface

- Responsive and user-friendly interface.
- Navigation menu with **Home, About, Dashboard, Story, Team, and Contact** pages.
- Embedded Tableau Dashboard and Story.
- Interactive filters, charts, maps, and tooltips.
- Accessible through any modern web browser without requiring Tableau Desktop.
- Easy to share and demonstrate using a public URL.

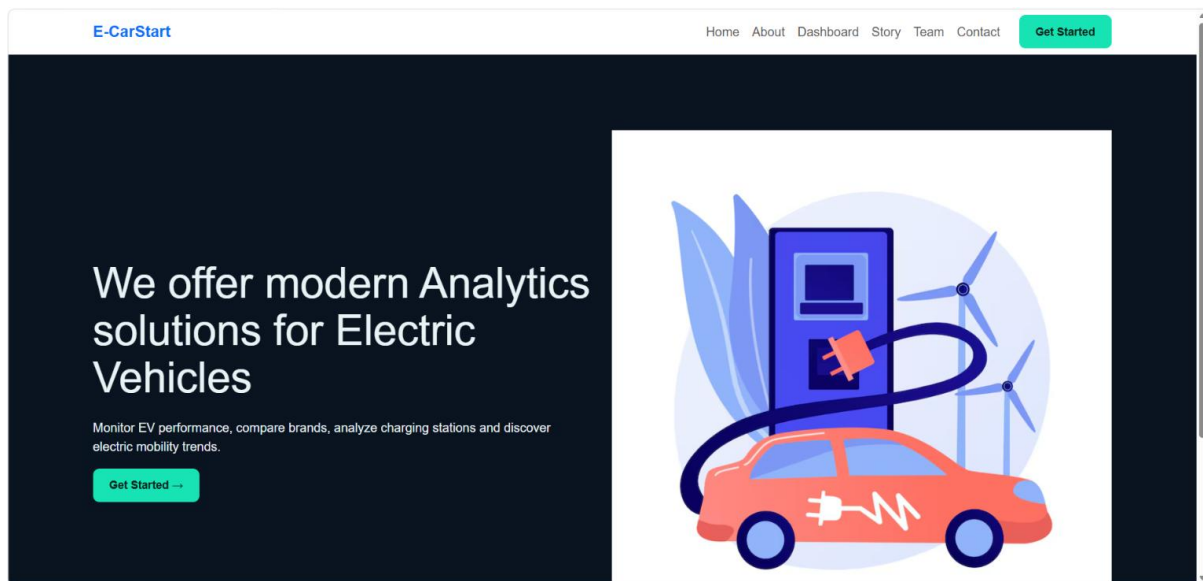


Figure 7.2.1: Home Page of the Flask Web Application

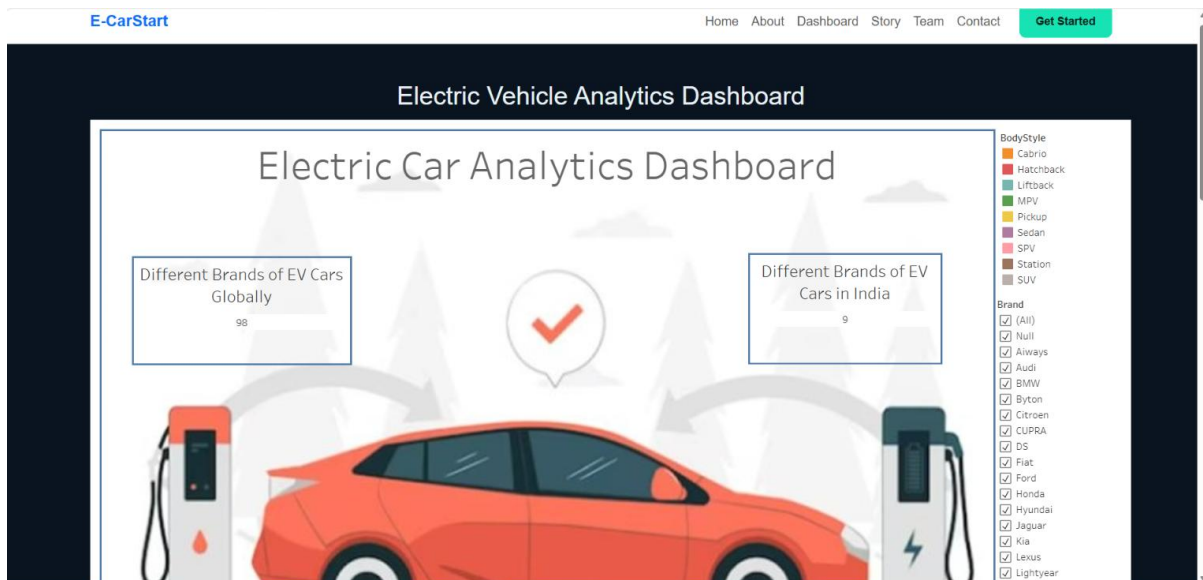


Figure 7.2.2: Embedded Tableau Dashboard

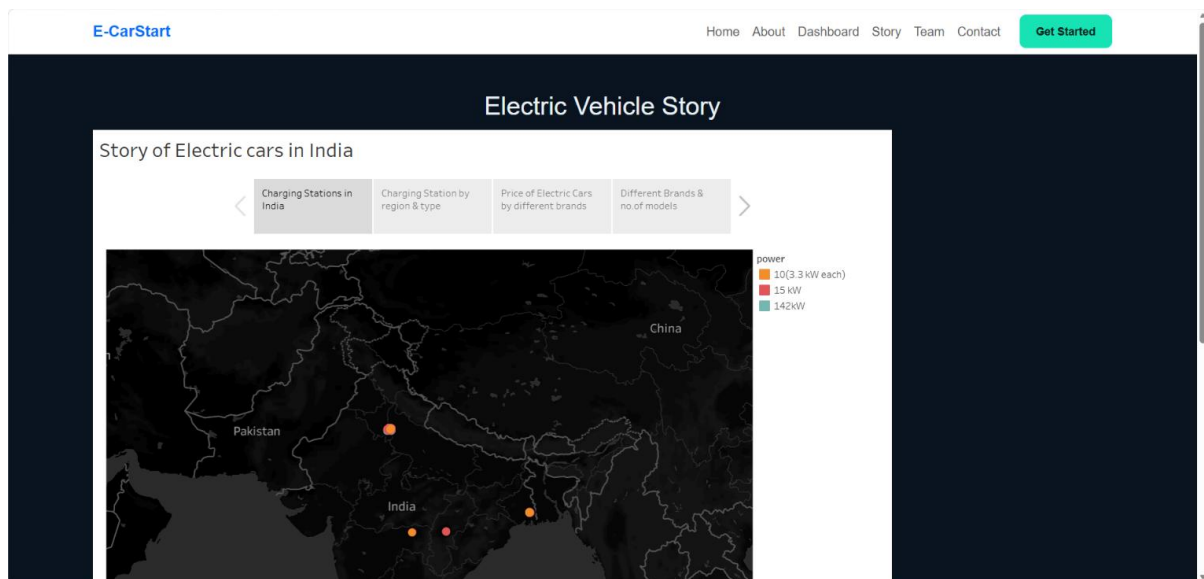


Figure 7.2.3: Embedded Tableau Story

Live Project:

<https://electric-vehicle-charge-and-range-pdbf.onrender.com>

Repository:

https://github.com/Meghana290607/Electric_Vehicle_Charge_and_Range_Analysis/tree/main

Results

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** was successfully developed using **MySQL, Tableau, and Flask**.

The EV dataset was stored and managed in MySQL, analyzed through Tableau, and presented using interactive dashboards and stories embedded in a Flask web application.

The project enables users to analyze electric vehicle charging infrastructure, battery performance, driving range, efficiency, top speed, pricing, and brand comparisons through interactive visualizations.

The implemented filters and calculated fields allow users to explore the data dynamically and obtain meaningful insights.

The published web application provides easy access to the dashboard and story through a user-friendly interface, ensuring a seamless analytical experience.

Advantages and Limitations

Advantages

The major advantages of the project are:

- Provides an interactive and user-friendly dashboard for EV data analysis.
- Enables quick comparison of different electric vehicle brands and models.
- Supports interactive filtering for customized analysis.
- Displays data using clear charts, maps, and graphical visualizations.
- Simplifies complex datasets into meaningful insights.
- Improves decision-making through visual analytics.
- Dashboard and Story are accessible through a web browser using Flask.
- Responsive design allows access from desktops, laptops, tablets, and mobile devices.
- Easy to maintain and update with new datasets.
- Supports sharing through Tableau Public and web deployment.

Limitations

Although the project provides valuable insights, it has a few limitations:

- The dashboard is based on a static dataset and does not include real-time data.
- Accuracy depends on the quality and completeness of the dataset.
- Internet connectivity is required to access the published dashboard.
- The analysis is limited to the available EV models and charging station data.
- Advanced predictive analytics and machine learning features are not included.
- The project currently focuses on visualization rather than real-time monitoring.

Future Enhancements

The project can be enhanced further by implementing the following features:

- Integration of real-time EV and charging station data.
- AI and Machine Learning models for predicting battery health and driving range.
- Route optimization based on charging station availability.
- Real-time charging station occupancy and availability.
- Mobile application development for improved accessibility.
- Cloud-based deployment for better scalability.
- Personalized EV recommendation system based on user preferences.
- Integration with IoT devices for live battery monitoring.
- Advanced analytics and forecasting for EV adoption trends.
- User authentication and personalized dashboards.

Conclusion and References

Conclusion

The **Visualization Tool for Electric Vehicle Charge and Range Analysis** successfully demonstrates the effective use of **Business Intelligence** techniques for analyzing electric vehicle data. By integrating **MySQL**, **Tableau**, and **Flask**, the project provides an interactive platform for exploring charging infrastructure, battery performance, driving range, vehicle efficiency, and pricing.

The dashboard and story transform complex datasets into meaningful visual insights, enabling users to compare different EV models and make informed decisions. The responsive Flask web application further enhances accessibility by allowing users to interact with the published dashboard through any web browser. Overall, the project highlights the importance of data visualization in supporting sustainable transportation planning and informed decision-making.

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